# The set of Dedekind cuts has the completeness property

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*Editorial note.* The counted statement is the author's own theorem labelled realcompleteness (915-917); the label sits on the begin line, so the statement span was computed from the label line and the proof span from the proof environment beginning at 919.

*Editorial note.* The author states the result in the compressed form that the set of cuts has the Completeness Property and spells the property out four lines earlier; that spelled-out version is retained as context so the counted statement has determinate content.

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit sets-functions-relations/arithmetization; the commented-out display at line 922 is retained verbatim inside the proof environment and is not active TeX.

*Editorial note.* The proof cites no other result, figure, file or bibliography entry and contains no non-ASCII character, so the package declares no external reference, no asset and no bibliography.

*Editorial scope.* One original result of the Open Logic Project unit sets-functions-relations/arithmetization, counted once: the set of cuts has the completeness property. Only the counted statement, its own author proof, the author's definition of a cut and the spelled-out form of the property are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

*Allowed standard prerequisites (editorial).*

- Cut. A cut is a subset alpha of the rationals that is non-empty and proper, initial (if p $<$ q and q is in alpha then p is in alpha) and has no greatest element; the reals are the cuts.. The proof verifies exactly these three clauses for the union of the family (923-936) and uses them for the members of the family when it passes from p $<$ q in the union to p in the union.
- Order on cuts. For cuts alpha and beta, alpha is at most beta exactly when alpha is a subset of beta.. Upper bound and least upper bound are read through this order in the second half of the proof (937-938).

- Union of a family of sets. The union of a family S is the set of p that belong to some member alpha of S.. The candidate supremum is defined as the union of S, and both halves of the proof unpack membership in it (920-921, 930-935, 938).

**Definition 0.1** (Cut)**.** A *cut* $\alpha$ is any non-empty proper initial segment of the rationals with no greatest element. That is, $\alpha$ is a cut iff:

1. *non-empty, proper*: $\emptyset \neq \alpha \subsetneq \mathbb{Q}$
2. *initial*: for all $p, q \in \mathbb{Q}$: if $p < q \in \alpha$ then $p \in \alpha$
3. *no maximum*: for all $p \in \alpha$ there is a $q \in \alpha$ such that $p < q$

Then $\mathbb{R}$ is the set of cuts.

As before, having defined some entities, we next need to define basic functions and relations upon them. We begin with an easy one:

$$\alpha \leq \beta \text{ iff } \alpha \subseteq \beta$$

This definition of an order allows to *state* the central result, that the set of cuts has the Completeness Property. Spelled out fully, the statement has this shape. If $S$ is a non-empty set of cuts with an upper bound, then $S$ has a least upper bound. In more detail: there is a cut, $\lambda$, which is an upper bound for $S$, i.e. $(\forall \alpha \in S)\alpha \subseteq \lambda$, and $\lambda$ is the least such cut, i.e. $(\forall \beta \in \mathbb{R})((\forall \alpha \in S)\alpha \subseteq \beta \rightarrow \lambda \subseteq \beta)$. Now here is the proof of the result:

**Theorem 0.2.** *The set of cuts has the Completeness Property.*

*Proof.* Let $S$ be any non-empty set of cuts with an upper bound. Let $\lambda = \bigcup S$. We first claim that $\lambda$ is a cut:

1. Since $S$ has an upper bound, at least one cut is in $S$, so $\emptyset \neq \lambda$. Since $S$ is a set of cuts, $\lambda \subseteq \mathbb{Q}$. Since $S$ has an upper bound, some $p \in \mathbb{Q}$ is absent from every cut $\alpha \in S$. So $p \notin \lambda$, and hence $\lambda \subsetneq \mathbb{Q}$.
2. Suppose $p < q \in \lambda$. So there is some $\alpha \in S$ such that $q \in \alpha$. Since $\alpha$ is a cut, $p \in \alpha$. So $p \in \lambda$.
3. Suppose $p \in \lambda$. So there is some $\alpha \in S$ such that $p \in \alpha$. Since $\alpha$ is a cut, there is some $q \in \alpha$ such that $p < q$. So $q \in \lambda$.

This proves the claim. Moreover, clearly $(\forall \alpha \in S)\alpha \subseteq \bigcup S = \lambda$, i.e. $\lambda$ is an upper bound on $S$. So now suppose $\beta \in \mathbb{R}$ is also an upper bound, i.e. $(\forall \alpha \in S)\alpha \subseteq \beta$. For any $p \in \mathbb{Q}$, if $p \in \lambda$, then there is $\alpha \in S$ such that $p \in \alpha$, so that $p \in \beta$. Generalizing, $\lambda \subseteq \beta$. So $\lambda$ is the *least* upper bound on $S$. □